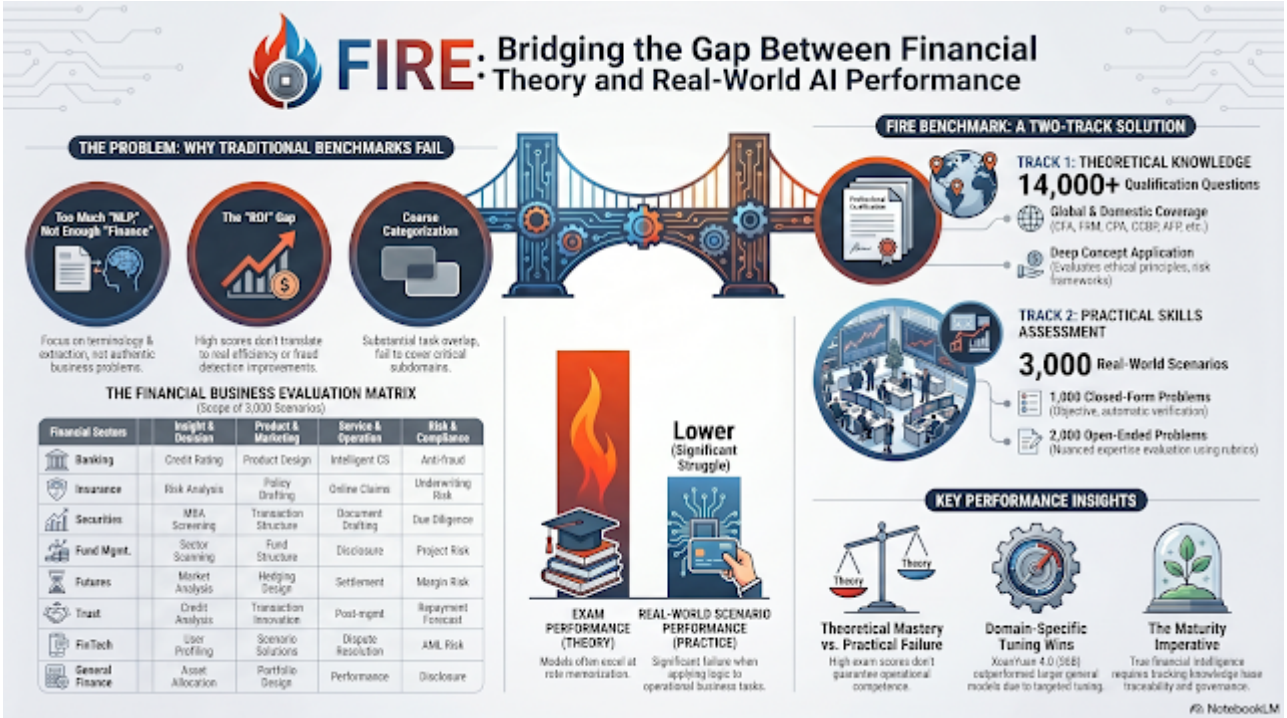
Financial Intelligence & Reasoning Evaluation (FIRE) × Governed Knowledge Objects

Toward a Maturation-Aware Evaluation Stack for Financial LLMs

0. Abstract

Financial LLM evaluation has improved, but it still suffers from a structural asymmetry. We are getting better at measuring whether a model can answer finance questions, yet we are still weaker at measuring the maturity, traceability, and governance quality of the knowledge substrate that supports those answers. FIRE is an important advance because it moves financial evaluation beyond shallow finance-flavored NLP and toward a benchmark that jointly measures theoretical knowledge and practical scenario reasoning through qualification questions, real-world business problems, matrix-based coverage, and rubric-based scoring for open-ended tasks. At the same time, the governed knowledge-object architecture makes a different but complementary move: it argues that persistent wiki pages should not be treated as flat prose artifacts, but as phase-specific knowledge objects that mature from source-grounded Raw Objects into perspective-bound Mature Objects under explicit trace, residual, and coverage governance.

This article argues that the next serious financial LLM stack should combine these two directions. FIRE tells us what to test. Governed knowledge objects tell us how to structure the knowledge base that is being tested. The result is a maturation-aware evaluation stack in which benchmark performance is no longer read as a property of the model alone, but as the joint outcome of model capability, object maturity, perspective discipline, residual honesty, and replayable assimilation history. In that sense, the proposal is not to replace FIRE, but to complete it.





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1. The Real Problem: Financial Intelligence Is Not Only Answer Quality

The central contribution of FIRE is that it refuses to confuse financial intelligence with surface fluency. The benchmark explicitly argues that earlier finance benchmarks often remain too close to conventional NLP tasks, too coarse in their categorization, and too weakly connected to actual business value. In response, FIRE splits evaluation into two parts: a large theoretical knowledge layer based on 14,000+ qualification-exam questions, and a practical layer built from 3,000 real-world financial scenarios, including 1,000 problems with reference answers and 2,000 open-ended problems scored through detailed rubrics and an automated scoring pipeline.

That is already a major correction. It says, in effect, that financial AI should be judged not only by whether it “knows finance,” but by whether it can operationalize financial knowledge inside business workflows such as insight generation, product design, service operations, and risk or compliance work. FIRE’s matrix-style design is important precisely because it refuses the comfort of narrow evaluation. It distributes questions across business functions and sectoral domains so that benchmark success becomes harder to fake through shallow memorization or generic reasoning strength alone.

But even that is not yet the whole problem. A model can underperform on a finance task for at least four different reasons:

1. the model lacks the necessary reasoning ability
2. the model lacks the necessary financial knowledge
3. the model has access to knowledge, but the knowledge base is immature or poorly governed
4. the runtime cannot tell which of the above actually caused failure

FIRE significantly improves diagnosis of the first two. The governed knowledge-object architecture helps with the third and fourth.

2. What FIRE Solves — and What It Deliberately Does Not Solve

FIRE solves an evaluation design problem. It gives the field a more serious target. Instead of asking whether a model can answer isolated finance questions, it asks whether a model can navigate the difference between textbook knowledge and real financial scenarios. Instead of relying only on multiple-choice scoring, it also introduces rubric-based grading for open-ended tasks, which is especially important in finance, where many practical answers are not reducible to one short canonical string.

Yet FIRE is still a benchmark, not a full knowledge runtime. It can tell us that a system scored poorly on a scenario. It can even tell us where the weakness clusters. But it does not, by itself, tell us whether the failure came from missing source grounding, premature concept flattening, hidden perspective mixing, erased residual conflict, stale object structure, or opaque mature-layer rewrites. Those are not weaknesses of FIRE. They are simply outside its intended boundary. FIRE measures capability. It does not claim to be a knowledge-maturation architecture.

That boundary matters. If we ask FIRE to answer questions it was not designed to answer, we will misread its scores. A low score may be interpreted as "the model is weak," when the deeper cause is "the knowledge substrate feeding the model has not matured honestly enough to support the task."

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3. Why Page-Centric Knowledge Is Not Enough for Finance

The governed wiki paper starts from a different diagnosis. It argues that even persistent wiki systems remain insufficient if they treat pages as flat prose files rather than staged knowledge objects. A page may be readable and useful, yet still leave unclear what was absorbed, what was rejected, what remains residual, and which perspective was active when the consolidation occurred. The paper therefore makes a decisive shift from page maintenance to object maturation. Its thesis is that source-compiled pages should function as immature concept objects designed for later universe-bound assimilation into mature concept objects, under explicit trace, residual, and coverage governance.

This distinction is especially important in finance. Financial knowledge is not merely a set of facts. It includes instruments, treatments, exceptions, reporting rules, risk logic, legal-adjacent constraints, numerical relationships, and domain-specific conflicts that often remain partially unresolved under one interpretation while becoming cleaner under another. A finance page that looks polished may still conceal fragile closure, mixed perspectives, or untracked exceptions. In practical systems, that produces a dangerous illusion: high readability, low governability.

The governed object architecture responds by introducing a phase ladder:

$$R \rightarrow O_raw \rightarrow O_mature \rightarrow O_ops. \quad (3.1)$$

If an inspirational layer is included, the richer form is:

$$R \rightarrow O_raw \rightarrow \{ O_mature, O_inspire \}. \quad (3.2)$$

Here R is the immutable raw source layer, O_raw is the source-grounded Raw Object layer, O_mature is the perspective-bound Mature Object layer, and O_inspire is the exploratory layer for weak signals, recurring near-misses, or unresolved but promising patterns. This is not merely

a storage trick. It is an honesty mechanism.

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4. The Missing Complement to FIRE: Maturity-Aware Evaluation

Once FIRE and governed objects are placed side by side, the complement becomes obvious.

FIRE asks: can the system answer finance questions and solve realistic finance scenarios? Governed objects ask: what kind of knowledge maturity made that answer possible, and how replayable was the path that produced it?

This suggests a broader evaluation unit:

Financial performance = f(model, objects, perspective, runtime, governance). (4.1)

A more explicit formulation is:

Perf_fin = F(M, K_mature(U), P, G). (4.2)

where:

M = the model
K_mature(U) = the mature knowledge layer under active universe U
P = the runtime protocol or task procedure
G = the governance surface, including trace, residual, and coverage discipline

This does not deny the importance of model quality. It says that in realistic finance systems, benchmark performance is usually not a pure property of M alone. It is co-produced by the maturity and honesty of the knowledge environment in which M operates.

That is where the object architecture adds something FIRE alone cannot add. It lets us distinguish between:

- model weakness
- knowledge weakness
- governance weakness
- perspective-mixing weakness
- replayability weakness

In financial contexts, that distinction is not cosmetic. It is the difference between “improve the model” and “repair the knowledge operating system.”

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5. A Proposed Maturation-Aware Evaluation Stack

The combined stack can be written compactly as:

K_mature(U) = Assimilate(U, K_raw, Σ, Tr, Res, Cov). (5.1)

This is the core maturation equation from the governed object architecture: raw material becomes mature knowledge only through perspective-bound assimilation under schema, trace, residual, and coverage governance.

Now add FIRE as the external performance surface:

$$E_{FIRE} = Bench(M, K_{mature}(U), T_{fin}). \quad (5.2)$$

where T_{fin} denotes the financial task family induced by FIRE: qualification questions, structured practical problems, and open-ended scenario tasks.

The real stack is therefore:

$$R \rightarrow O_{raw} \rightarrow O_{mature}(U) \rightarrow \text{task runtime} \rightarrow \text{FIRE evaluation} \rightarrow \text{diagnostic feedback}. \quad (5.3)$$

Or, if we include the offline assimilation cycle explicitly:

day loop = ingest / query / light lint. (5.4)
night loop = assimilation / reconciliation / coverage consolidation. (5.5)

The paper on governed objects calls this bounded offline layer the Perspective Assimilation Engine, or PAE:

$$PAE(U, B_t, M_t) \rightarrow (M_{(t+1)}, Cov_{(t+1)}, Res_{(t+1)}, Tr_{(t+1)}). \quad (5.6)$$

Its purpose is to revisit Raw Objects in batches, under one active universe per pass, and perform broader consolidation while preserving coverage updates, residual packets, and replayable trace.

This gives us a complete improvement loop:

1. ingest finance sources into Raw Objects
2. mature them under a declared finance universe
3. run financial tasks drawn from FIRE
4. identify failures not only by score but by coverage, residual, and assimilation state
5. feed those diagnostics back into the next assimilation cycle

That is what a maturation-aware evaluation stack means in practice.

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6. New Metrics That FIRE Could Be Paired With

FIRE already provides a strong task-facing surface. But if it is paired with governed objects, the evaluation family becomes richer. We no longer ask only whether the answer was good. We also ask whether the knowledge state that produced the answer was mature enough to deserve trust.

A practical metric family might be:

$$Score_{total} = \alpha \cdot Score_{FIRE} + \beta \cdot Cvg + \gamma \cdot MPrec + \delta \cdot RHon + \varepsilon \cdot Rpl - \zeta \cdot Drift. \quad (6.1)$$

where:

$Score_{FIRE}$ = benchmark performance on FIRE tasks
Cvg = coverage quality
$MPrec$ = merge precision
$RHon$ = residual honesty
Rpl = replayability
$Drift$ = mature-layer drift or instability

The governed object paper explicitly motivates these dimensions. Coverage tells us what was absorbed. Residual tells us what remains outside closure. Assimilation events tell us how the current state came to exist. Without these, a page may look “better” while the architecture becomes less honest and less replayable.

A finance-oriented interpretation of these metrics could look like this:

- **Coverage quality**: did the system actually absorb the relevant source segments into the right mature objects?
- **Merge precision**: did it merge concepts correctly, or did it over-collapse distinct treatments, instruments, or mechanism types?
- **Residual honesty**: did it preserve unresolved exceptions, ambiguities, and conflicts, or did it prematurely smooth them away?
- **Replayability**: can a human reviewer reconstruct why the system now believes what it believes?
- **Drift boundedness**: is the mature finance layer changing in a controlled way, or being silently rewritten by every new ingestion event?

These are not replacements for FIRE. They are the missing internal diagnostics that make FIRE results more interpretable.

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7. Why Finance Especially Needs This Combination

Finance is an unusually demanding environment for LLM systems because it sits at the intersection of theory, formal structure, operational timing, regulatory pressure, exception handling, and domain-sensitive language. In such domains, the question is rarely only “is the answer correct?” The more serious question is often: “under what perspective was this answer formed, what unresolved remainder still exists, and can the path be reviewed later?”

That is exactly why the object architecture’s insistence on single-perspective execution is so useful. A Raw Object may be eligible for several universes, but each assimilation pass should declare one active universe only. In the language of the paper:

multi-home eligibility, single-perspective execution. (7.1)

This matters because hidden perspective mixing is cheap now and expensive later. In finance, silent blending of accounting, mechanism, legal, and business perspectives often creates polished but unstable answers. Governed maturity avoids this by requiring declared perspective during consolidation.

The same is true of residuals. In many finance tasks, residual is not a sign of failure. It is a sign that the system is honest enough not to pretend that a weakly grounded exception, perspective clash, or unresolved treatment issue has already been settled. That is not a bug. It is often the right epistemic posture.

So the reason finance benefits especially from the combination is simple: finance punishes hidden immaturity.

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8. A Rollout Ladder for Real Systems

One of the best features of the governed-object architecture is that it does not force maximal complexity from day one. It explicitly proposes a rollout ladder in which additional machinery is added only when the current layer can no longer govern staleness, drift, or dishonesty honestly.

That principle can be translated directly into a FIRE-integrated deployment ladder.

Tier 1 — Benchmark-first

Use FIRE on a base model or finance-tuned model with a conventional retrieval or static knowledge layer. This reveals the external capability boundary.

Tier 2 — Raw Object layer

Convert the finance corpus into source-grounded Raw Objects with stable segments, source refs, and residual placeholders. This already improves recoverability and future diagnosability.

Tier 3 — Mature finance universe

Add one finance universe with explicit mature objects, one-active-universe assimilation discipline, and segment-level coverage. This upgrades the knowledge layer from readable storage to governed maturity.

Tier 4 — Night-time assimilation and feedback

Add a bounded PAE and close the loop with FIRE task failures. At this point, benchmark results begin to drive targeted knowledge maturation rather than generic reindexing.

Tier 5 — High-reliability governance

Add human arbitration at doctrine-sensitive boundaries, mature/inspirational split, domain packs, richer indices, and drift controls. This is the enterprise-grade regime.

This ladder matters because the proposal here is not “build a giant architecture.” The proposal is “match your evaluation and knowledge maturity to the actual pain profile of the system.”

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9. The Deeper Architectural Payoff

The deeper gain from combining FIRE with governed knowledge objects is not just better scores. It is better causal legibility.

Without the combination, a system failure on a finance scenario often yields a vague verdict:

the model failed. (9.1)

With the combination, the verdict can become much sharper:

the model failed because the relevant mature object was under-covered, the exception segment remained residual, the assimilation pass was performed under the wrong universe, and the prior merge decision was only provisional. (9.2)

That is a radically better engineering sentence.

It means financial AI can be debugged as a joint system rather than judged as a monolith. It also means progress can be targeted more intelligently. Sometimes the answer will be “train a better

model.” But sometimes the real answer will be “repair the knowledge object pipeline.”

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10. Conclusion

FIRE is one of the clearest recent advances in financial LLM evaluation because it moves the field away from shallow finance-NLP proxies and toward a benchmark that jointly measures theoretical knowledge and practical scenario reasoning. The governed knowledge-object architecture makes a different but equally important move: it argues that persistent knowledge must be treated as a staged maturation process rather than a pile of polished pages. These two moves are not redundant. They solve different halves of the same larger problem.

The synthesis proposed in this article is therefore straightforward:

FIRE tells us whether a financial LLM system performs well.
Governed knowledge objects tell us what kind of knowledge maturity made that performance possible. (10.1)

Put differently:

benchmark quality without knowledge maturity is hard to diagnose;
knowledge maturity without benchmark pressure is hard to validate. (10.2)

A serious financial LLM stack should therefore aim for both. It should benchmark the model against realistic financial tasks, but it should also know what phase its knowledge artifacts are in, what perspective was active, what has been assimilated, what remains residual, and how the mature layer came to exist. That is the point at which financial evaluation stops being only a test suite and becomes part of a real knowledge operating system.

Reference

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By Xiyuan Zhang, Huihang Wu, Jiayu Guo, Zhenlin Zhang, Yiwei Zhang, Liangyu Huo, Xiaoxiao Ma, Jiansong Wan, Xuewei Jiao, Yi Jing, Jian Xie, 2026

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I am merely a midwife of knowledge.